

NC(SMC)² 2026 Team Round

NC(SMC)² Problem Writers

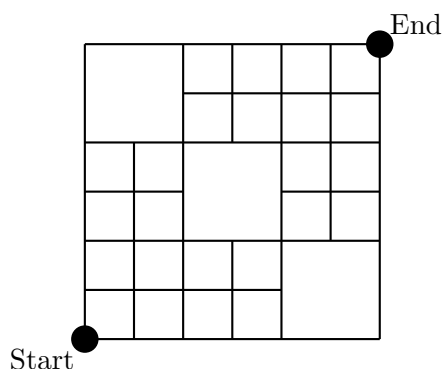
May 9th, 2026

1. Aaron wakes up at 8:05 for his 8:30 class. If he spends 5 minutes getting out of bed, 3 minutes brushing teeth, 2 minutes changing clothes, and 8 minutes preparing his backpack, how many minutes does he have left until class starts?
2. Ethan is designing a robot. If he spends eight hours a day sleeping, six hours per day in school, and half of his remaining time designing his robot, how many hours does he spend designing a robot each week?
3. Clark's elementary school has created a new rule randomizing the number of classes he has each day. Each school day, Clark is randomly scheduled to take either 27, 3, or 0 classes. What is the expected number of classes he will take in 90 school days?
4. "Little" Ica is the fattest pegasus in Amphoreus. On Sunday, she eats 10 pounds of food. Each day, she eats 6 more pounds of food compared to the previous day. After the next Sunday, how many pounds of food has she eaten?
5. Let $a_0 = 5$ and define $a_{i+1} = a_i^2 - a_i$ for $i \geq 0$. What are the last two digits of a_{2026} ?
6. Out of Andrew, Bryan, Cantrell, David, and Erik, some of them always tell the truth and others always lie. Erik, as your friend, is telling the truth. The other four each tell you:
 - Andrew says that Bryan is lying.
 - Bryan says that David is lying.
 - Cantrell says that David is telling the truth.
 - David says that Erik is lying.

How many people are telling the truth (including Erik)?

7. Colin needs to get from the cafeteria on the ground (0th) floor to his science classes on the higher floors. There are two options: take the stairs or take the elevator. Colin can run up one flight of stairs in 10 seconds. Meanwhile, the elevator takes 20 seconds to come down to ground level and 5 seconds to travel up one floor. What is the lowest floor number where it takes less time to use the elevator?

8. Alan the genie grants orzness to people. Every day, when you visit him, you have a $\frac{6}{7}$ chance of becoming orz. The probability that you are orz after visiting him for 3 days can be written as $\frac{m}{n}$ in simplest form. What is $m + n$?
9. Lucas is buying a birthday cake. The cake costs \$0.05 per cubic centimeter and icing costs \$0.04 per square centimeter, and an additional \$3 for icing lettering. Lucas decides on a cake with a square base and a height of 4 centimeters, with icing on the top and all four sides, and icing lettering. The cake costs \$23.48 before tax. What is the side length of the base of his cake in centimeters?
10. Nelson writes one problem every four minutes. Alan deletes a problem every five minutes. They both start at the same time, with an empty test. After how many minutes will there first be 20 problems be on the test?
11. Phantylia starts with n flowers. She first gives $\frac{1}{2}$ of her flowers to Nanook. Then, she gives $\frac{1}{6}$ of the remaining flowers to Yaoshi. Finally, she burns $\frac{1}{10}$ of the flowers and is left with an integer amount of flowers. What is the minimum amount of flowers Phantylia started with?
12. Carter is playing tic tac toe on a 5×3 grid. A player wins when they place three of their own pieces in a row next to each other horizontally, vertically, or diagonally. Count the number of ways to place three pieces such that Carter wins.
13. Kevin is driving in New York City in November. However, due to the Macy's Thanksgiving Day Parade certain blocks are closed. How many different paths of the shortest possible distance can Kevin take from his hotel to the coffee shop?



14. There are two points where the graphs of the following points intersect.

$$x^2 + (y - 5)^2 = 25$$

$$(x - 4)^2 + y^2 = 16$$

The sum of the x -coordinates of the intersection points is $\frac{m}{n}$ when expressed in simplest form. Compute $m + n$.

15. What is the largest number that cannot be the smallest side in a Pythagorean triple (right triangle with 3 integer side lengths)?
16. A positive integer is written at each vertex of a square. On each side of the square, the product of the numbers at the endpoints of the side is written. The sum of the four numbers on each side of the square is 77. What is the sum of the numbers on the vertices of the square?
17. There are 6 points evenly spaced around the circle. Each point is paired with another point, and three lines are drawn according to this pairing. The probability that no lines intersect each other can be written as $\frac{m}{n}$ in simplest form. What is $m + n$?
18. Colin cuts out a regular hexagon and three circles of radius 12. If the hexagon can be completely covered by the three circles, compute the largest possible side length of the hexagon.
19. Consider the sequence $1^1, 2^2, 3^3, \dots$. What is the period of the remainders of this sequence when divided by 67?
20. A mathematical game is played between two very clever students, A and B . The referee first writes the integers

$$2, 3, 4, \dots, 2026$$

on the blackboard, and then randomly erases one of them. After that, B and A alternately erase one number from those remaining, with B moving first, then A , and so on. When only two numbers are left, if those two numbers are relatively prime, then A wins; otherwise, B wins.

According to these rules, the probability that A wins is $\frac{p}{q}$. Find $p + q$.